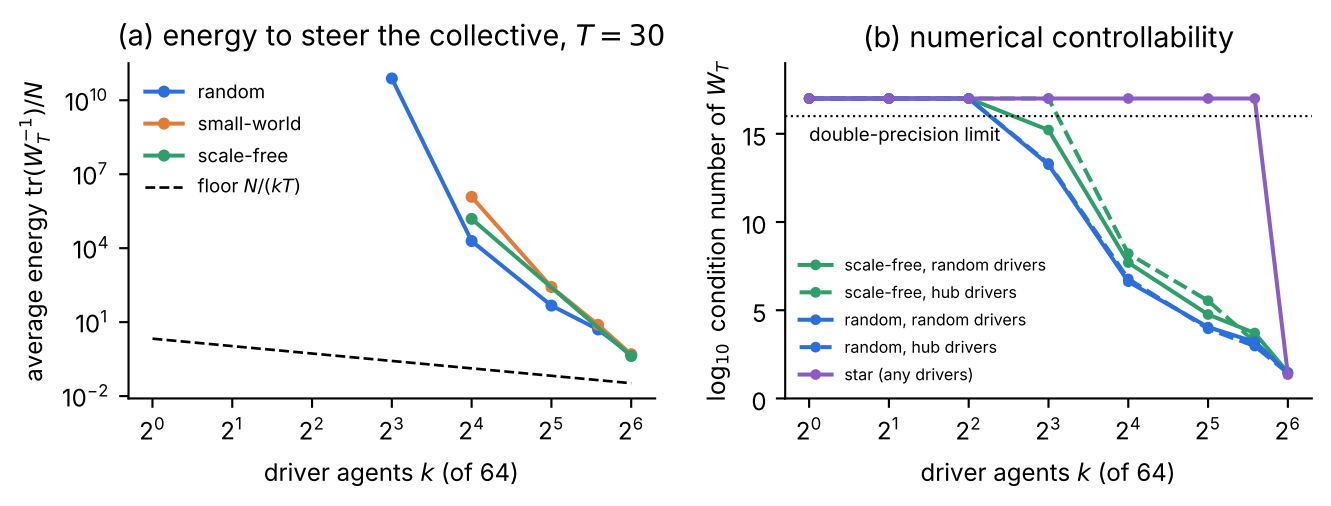
(a) energy to steer the collective, $T = 30$
average energy $\mathrm{tr}(W_T^{-1})/N$
10^{10}
10^{7}
10^{4}
10^{1}
10^{-2}
random
small-world
scale-free
floor $N/(kT)$
2^0 2^1 2^2 2^3 2^4 2^5 2^6
driver agents k (of 64)
(b) numerical controllability
$\log_{10}$ condition number of W_T
15
10
5
0
double-precision limit
scale-free, random drivers
scale-free, hub drivers
random, random drivers
random, hub drivers
star (any drivers)
2^0 2^1 2^2 2^3 2^4 2^5 2^6
driver agents k (of 64)